

A - Min of Sum of XOR

分值: 500 分

Problem Statement

You are given a positive integer N .

给定一个正整数 N 。

Find one permutation $P = (P_1, P_2, \dots, P_N)$ of $(1, 2, \dots, N)$ that minimizes $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j$.

找到一个 $(1, 2, \dots, N)$ 的排列 $P = (P_1, P_2, \dots, P_N)$, 使得 $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j$ 最小。

Here, $\bigoplus_{1 \leq j \leq i} P_j$ is defined as the bitwise XOR of $P_1, P_2, \dots, P_i$.

这里, $\bigoplus_{1 \leq j \leq i} P_j$ 的定义是 $P_1, P_2, \dots, P_i$ 的按位 XOR。

You are given T test cases; solve each of them.

给定 T 组测试用例, 每组测试用例都需要解决。

What is bitwise ?

什么是按位 ?

The bitwise XOR of non-negative integers A and B , $A \oplus B$, is defined as follows.

非负整数 A 和 B 的按位 XOR 运算结果 $A \oplus B$ 定义如下。

- In the binary representation of $A \oplus B$, the digit at the 2^k ($k \geq 0$) place is 1 if exactly one of the digits at the 2^k place in the binary representations of A and B is 1, and 0 otherwise.
- 在 $A \oplus B$ 的二进制表示中, 若 A 和 B 的二进制表示中第 2^k 位的数字恰好有一个是 1, 则第 2^k ($k \geq 0$) 位的数字为 1, 否则为 0。

For example, $3 \oplus 5 = 6$ (in binary: $011 \oplus 101 = 110$).

In general, the bitwise XOR of k non-negative integers $p_1, p_2, p_3, \dots, p_k$ is defined as $(\dots((p_1 \oplus p_2) \oplus p_3) \oplus \dots \oplus p_k)$, and it can be proved that this does not depend on the order of $p_1, p_2, p_3, \dots, p_k$.

例如, $3 \oplus 5 = 6$ (二进制表示为: $011 \oplus 101 = 110$)。一般来说, k 个非负整数 $p_1, p_2, p_3, \dots, p_k$ 的按位 XOR 运算结果定义为 $(\dots((p_1 \oplus p_2) \oplus p_3) \oplus \dots \oplus p_k)$, 可以证明该结果与 $p_1, p_2, p_3, \dots, p_k$ 的运算顺序无关。

Constraints

- $1 \leq T \leq 10^3$
- $1 \leq N$
- The sum of N over all test cases is at most 2×10^5 .
- All input values are integers.
- $1 \leq T \leq 10^3$
- $1 \leq N$
- 所有测试用例的 N 之和不超过 2×10^5 。
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出:

```
T
case1
case2
⋮
caseT
```

Each test case is given in the following format:

每组测试用例按照以下格式给出:

N

Output

Output the answers for the test cases in order, separated by newlines.

按顺序输出所有测试用例的答案，答案之间用换行分隔。

For each test case, output a permutation $P = (P_1, P_2, \dots, P_N)$ of $(1, 2, \dots, N)$ that minimizes $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j$, separated by spaces.

对于每组测试用例，输出一个使得 $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j$ 最小的 $(1, 2, \dots, N)$ 的排列 $P = (P_1, P_2, \dots, P_N)$ ，排列中的元素用空格分隔。

If there are multiple permutations P that achieve the minimum, any of them will be accepted.

如果存在多个达到最小值的排列 P ，输出其中任意一个即可。

Sample Input 1

```
3
3
1
7
```

Sample Output 1

```
1 3 2
1
4 5 3 2 6 7 1
```

Consider the first test case.

考虑第一组测试用例。

If $P = (1, 3, 2)$, then $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j = 1 + (1 \oplus 3) + (1 \oplus 3 \oplus 2) = 1 + 2 + 0 = 3$.

如果 $P = (1, 3, 2)$, 则 $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j = 1 + (1 \oplus 3) + (1 \oplus 3 \oplus 2) = 1 + 2 + 0 = 3$.

The value of $\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j$ cannot be made less than 3, so outputting $P = (1, 3, 2)$ is correct.

$\sum_{i=1}^N \bigoplus_{1 \leq j \leq i} P_j$ 的值无法小于 3, 因此输出 $P = (1, 3, 2)$ 是正确的。

Outputting $P = (2, 3, 1)$ is also correct.

输出 $P = (2, 3, 1)$ 也是正确的。